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WEB APPLICATION USING MERN STACK
Project review

SYNCSTAR – Real-Time Multi-Game

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PRESENTATION OUTLINE

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ABSTRACT

SyncStar is a real-time collaborative platform built using the MERN stack and Socket.io.

It enables users to create instant private rooms without login and collaborate through:

- Whiteboard drawing
- Live chat
- User presence tracking

The system uses **WebSockets** for low-latency communication and **MongoDB** for data storage, ensuring smooth and scalable performance.

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INTRODUCTION

- Growing need for real-time collaboration tools
- Existing tools require login and setup
- SyncStar provides instant access without barriers
- Built using React, Node.js, Express, and MongoDB
- Focus on speed, simplicity, and accessibility

PROBLEM STATEMENT

Traditional tools:

- Require account creation
- Have complex setup
- Cause delay in quick collaboration

Problem:

No lightweight platform for instant collaboration

Solution:

SyncStar provides instant room creation without login

SCOPE OF THE PROJECT

- Full-stack web application using MERN stack
- Real-time collaboration using WebSockets
- Works across devices (responsive design)
- Supports multiple users in one room
- Designed for education and teamwork

MODULES

- Room Management Module
- Whiteboard Module
- Live Chat Module
- User Presence Module
- Each module works together for real-time collaboration.

DATABASE DESIGN

Database: MongoDB

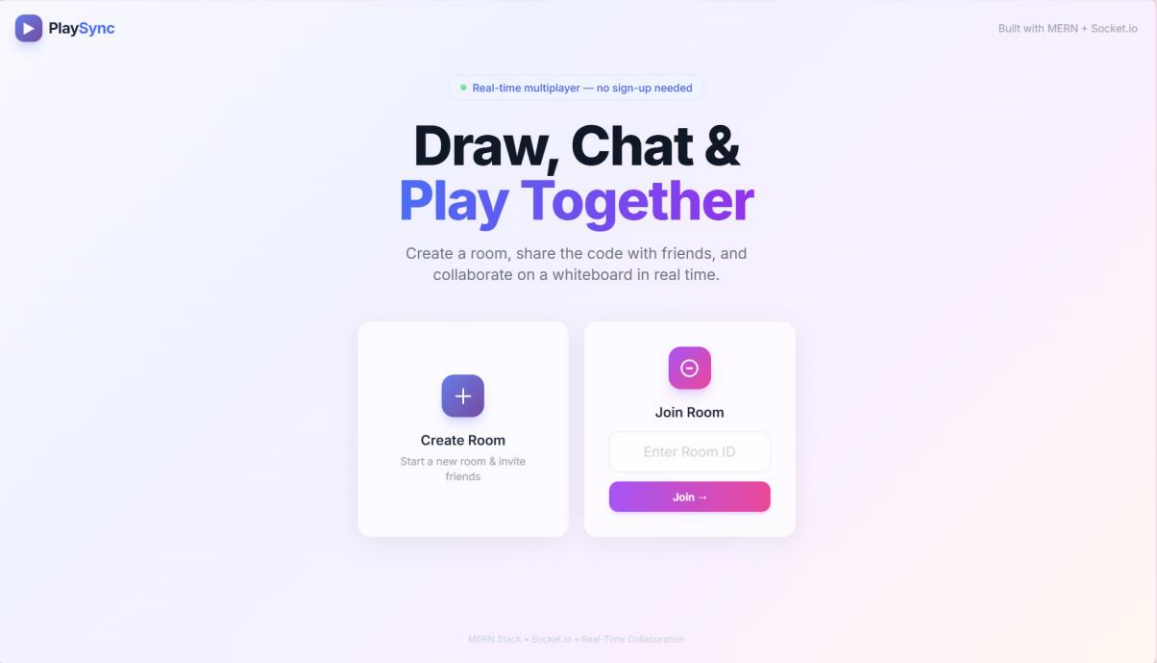
Collections:

- **Rooms**
    - roomId
    - createdAt
    - isActive
- **Chats**
    - roomId
    - userName
    - message
    - timestamp

Ensures data persistence and fast access.

OUTPUT SCREENSHOTS

Home Screen (Create/Join Room)



Whiteboard Screen

Chat Interface

CONCLUSION

- Successfully built a real-time collaboration platform
- Eliminates login barriers
- Supports instant communication and drawing
- Uses efficient MERN + WebSocket architecture
- Provides scalable and user-friendly solution

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REFERENCES

- Node.js Documentation
- React Documentation
- MongoDB Documentation
- Socket.io Documentation

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THANK YOU